

Correlation Witnesses versus Magic: A k -Resolved Nonstabilizerness Map for $SU(2)_k$ Anyon Fusion Spaces

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Standard correlation witnesses — spatial Bell/CHSH inequalities, temporal Leggett–Garg K_3 , and KCBS contextuality — are known to detect nonstabilizerness (“magic”) in some settings. We report a worked example, a k -resolved nonstabilizerness map of the $SU(2)_k$ anyon braid-representation family, in which a standard temporal witness is instead systematically *blind*: at $k = 4$ the three-time Leggett–Garg witness saturates its macrorealistic bound exactly ($K_3 = 1.000000$ over every state, braid element, and measurement axis, in the equal-time-step protocol of Table I with $V_1=V_2$) while the single-qubit fusion channel carries near-maximal nonstabilizerness ($M_2 = 0.5585$, 95% of the finite-dimensional ceiling $\log_2(3/2)$). We prove this blindness as a structural theorem: the witness depends only on the Bloch-sphere Gram geometry of the braid orbit, not on nonstabilizerness, and $k = 4$ happens to align the fixed measurement axis with a threefold orbit symmetry (Bloch dot products all $-1/3$) that caps the witness at 1; a finite, Clifford-generating group — the chiral octahedral group, the $k=2$ braid image — with a misaligned axis reaches $K_3 = 3/2$ under a two-propagator protocol. Neither finiteness of the braid image nor the Clifford property is by itself the mechanism. We complement this temporal certificate with three independent certificates of genuine nonstabilizerness — two of them long-range (a doubled-Fibonacci mutual-information witness, $H = 1.700979$, and a gauge-invariant minimum ground-space stabilizer Rényi entropy, ≥ 6.5 against an exact-zero toric-code control), the third a complementary gate-based non-Cliffordness measure — an honest negative control confirming that leakage-free constructions remain strictly additive, and a hardware-oriented single-qubit signal-to-noise prediction. We further probe the $k = 4$ dissociation at its shared $d = 3$ interface with Kochen–Specker–Klyachko contextuality, where the KCBS witness shows only a weak, nongeneric correlation with magic. Two reconciliations with the literature are made explicit: the present result contradicts neither the equivalence of maximal nonlocality and maximal magic established for optimized magic states in a different game, nor the contextuality-supplies-magic theorem; both concern a different object than the fixed, per- k braid generator studied here.

I. INTRODUCTION AND RELATED WORK

Nonstabilizerness, or “magic,” is the resource that lifts Clifford circuits to universal quantum computation [1, 2], and it is by now well established that standard Bell-type correlation witnesses can *detect* it: Macedo *et al.* [3] and Cusumano *et al.* [4] both construct Bell inequalities whose violation certifies nonstabilizerness, and the Ising-anyon no-go of Howard and Vala [5] shows, in the opposite direction, that Clifford-only (Ising) braiding transformations cannot violate a Bell inequality at all. Both directions concern whether, and how, a correlation witness sees magic. A companion paper realizes exactly the Howard–Vala Ising setting microscopically, on a concrete Kitaev honeycomb Hamiltonian, showing that the non-commuting CHSH Bell-operator *algebra* is realized locally while a genuine physical violation is not [26]; the magic resource whose presence or absence that companion paper’s argument turns on is exactly the subject of the present paper, and we cross-reference it bidirectionally throughout.

This paper reports the complementary direction, a

blind spot: a standard correlation witness — the temporal Leggett–Garg inequality $K_3 \leq 1$ [8, 9] — fails to detect near-maximal single-qubit magic present in a specific, structurally identified member of the $SU(2)_k$ braid-representation family, at $k = 4$. This is not a new claim that correlation witnesses can be blind to magic *in general* — Howard and Vala already establish an instance of witness blindness for Ising braiding [5] — but a k -resolved, numerically exhibited, and structurally explained dissociation map across the $SU(2)_k$ family, together with three independent certificates of genuine nonstabilizerness (two long-range, the third a complementary gate-based measure) and an honest negative control. We stress at the outset what this paper is *not*: it is not a claim that magic is generically decoupled from Bell/CHSH correlations — Macedo and Cusumano show the opposite in their respective constructions — and the present $k = 4$ result is the complementary blind spot on a single, finite braid generator, not a general no-go. A related but distinct construction, Robin and Savage’s anti-flatness witness of nonlocal magic in two-particle scattering [16], cites the same spatial magic–CHSH suppression as Macedo and Cusumano but treats a different Hilbert space (a two-qubit scattering channel, not anyons or an $SU(2)_k$ family) and does not touch the temporal or k -resolved question studied here; we note it as related work

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rather than a competing claim.

Two further reconciliations, closer to the present claim than the detect/blind distinction above, are necessary before we can state the $k = 4$ result without ambiguity. Howard [6] proves an *equivalence* between maximum nonlocality and maximum magic in a qudit CHSH-type game optimized over the magic *state*: this sounds directly opposed to any witness/magic dissociation. The distinction is the object being varied. Howard’s equivalence concerns the optimal single-qudit magic *state* in a fixed correlation game; the present $k = 4$ result concerns a fixed *braid generator* (no state optimization) at one specific value of the family index k , evaluated as a witness of whichever magic that braid generator’s fusion channel happens to carry. The two are different optimization problems over different objects, and neither result constrains the other. Second, Howard, Wallman, Veitch, and Emerson [7] prove that contextuality is necessary for the magic-state resource that promotes stabilizer computation to universality (odd-prime local dimension). We report below (Sec. VIII) a near-zero, nongeneric correlation between a KCBS contextuality witness and magic at $k = 4$; this is not in tension with the HWVE theorem. HWVE is a global existence/threshold statement about universality (odd-prime local dimension), whereas our KCBS correlation is a pointwise measurement over one finite- k orbit, not a threshold claim. The separate blindness of our temporal K_3 witness (Secs. III–IV) is of a different character still: K_3 is a temporal witness, not a spatial/polytope contextuality inequality, and the $k = 4$ dissociation is a structural corollary of a single braid generator being a z -axis rotation, not a general statement about contextuality and magic.

The rest of this paper is organized as follows. Section II fixes five nonstabilizerness axes and the $\log_2$ convention used throughout. Section III presents the k -resolved dissociation map, Fig. 1, with the $k = 4$ blind spot as the central result. Section IV proves the blindness as a structural theorem. Section V reports three independent certificates of nonstabilizerness — two long-range and a complementary gate-based axis. Section VI reports an honest negative control. Section VII gives a single-qubit hardware signal-to-noise prediction. Section VIII maps the dissociation onto an interface with KCBS contextuality, left open for a forthcoming companion analysis, and states the scope limitations of this work.

II. FIVE NONSTABILIZERNESS AXES

We fix one convention throughout: all magic monotones are reported in base-2 logarithm (bits), following the resource-theoretic construction of Veitch *et al.* [1], $M(\rho) = \log_2(1 + 2\text{sn}(\rho))$, where $\text{sn}(\rho)$ is the sum-negativity of the discrete Wigner representation of ρ . Byles, Forbes, and Pachos [13] report the qutrit gate-magic value $M_2 = \log(16/13)$ without specifying the logarithm base; in base 2 this is $M_2 = \log_2(16/13) = 0.2996$,

the value we cite below. The five axes evaluated across the $\text{SU}(2)_k$ family are: (1) the stabilizer 2-Rényi entropy (SRE) M_2 [17], proven to be a genuine magic monotone for Rényi index $\alpha \geq 2$ by Leone and Bittel [18], of the single-qubit braid-generator fusion channel (state magic); (2) the robustness of magic (RoM) [2]; (3) the amortized, gate-based non-Cliffordness of the braid generator itself (Sec. III); (4) the two independent long-range-nonstabilizerness certificates of Sec. V (which additionally re-presents axis (3) as a complementary third non-Cliffordness check); and (5) the temporal Leggett–Garg witness K_3 and the KCBS contextuality witness of Sec. VIII, both used as correlation witnesses rather than magic monotones. An independent, sampling-based route to the same family of monotones is Bell sampling [19], which we do not use directly but note as an alternative, hardware-compatible estimator.

III. THE k -RESOLVED DISSOCIATION MAP

For each k we consider the standard $\text{SU}(2)_k$ braid generator acting on a single fusion-channel qubit as a z -axis rotation by the level-dependent angle $\theta_k = \pi(k + 4)/(k + 2)$ [10, 11]; the braid image is dense in $\text{SU}(2)$ for every $k \geq 2$ with $k \notin \{2, 4, 8\}$ [12] (density alone, not finiteness, is the relevant classification for the “dense” rows of Table I below). We evaluate, independently, a correlation witness (the temporal Leggett–Garg K_3) and three magic monotones (the stabilizer 2-Rényi entropy M_2 , the robustness of magic RoM, and the amortized gate-based non-Cliffordness of Sec. IV) across $k = 2, \dots, 10$. Table I and Fig. 1 report all four quantities; the three magic monotones, evaluated by an independent re-implementation cross-checked by two further computations sharing verified conventions, agree exactly (Clifford floor 0 at $k = 2$, a finite non-Clifford plateau at $k = 4, 6, 8$, and the dense-representation ceiling $\log_2(3/2) = 0.585$ at $k = 3, 5, 7, 9, 10$), and the two K_3 columns (state- and axis-optimized K_3^{opt} , versus the fixed-measurement-axis value $K_3(Q=\hat{z})$ used operationally in Sec. IV) agree everywhere except at $k = 8$, where the two differ by construction (Sec. IV, scope remark).

The central object of this paper is the $k = 4$ column of Table I: the Leggett–Garg witness is pinned at its macrorealistic bound, $K_3 = 1.000000$, over *every* state, every braid word, and every measurement axis in the equal-time-step protocol ($V_1=V_2$). The axis-sweep is a genuine numerical search, not a fixed-axis artifact — the same axis-optimization protocol applied at $k = 8$ (Sec. IV) finds a strictly larger value there, 1.427 against the fixed-axis 1.342, so the search demonstrably explores nonaligned axes and still finds no $k = 4$ violation. The closed-form Gram-geometry argument of Sec. IV independently explains *why* the physical, fixed measurement axis used throughout this paper gives exactly 1 at $k = 4$ (it is aligned with a threefold orbit symmetry). In the

TABLE I. k -resolved dissociation map. FLW: dense in $SU(2)$ (Freedman–Larsen–Wang universality); for $k \geq 2$, finite braid images occur only at $k \in \{2, 4, 8\}$ [10], corresponding to the binary polyhedral groups $2O, 2T, 2I$ of orders 48, 24, 120 (McKay correspondence with the E_7, E_6, E_8 Dynkin diagrams). K_3^{opt} : Leggett–Garg witness in the equal-time-step protocol ($V_1=V_2$), maximized over states, braid words, and measurement axis. $K_3(Q=\hat{z})$: same witness at the fixed measurement axis used in Sec. IV. M_2 : stabilizer 2-Rényi entropy (state magic, $\log_2$). RoM: robustness of magic. All values reproduced by an independent re-implementation and cross-checked by two further evaluations sharing verified conventions; $|\Delta| \leq 10^{-6}$ throughout.

k	finite image	K_3^{opt}	$K_3(Q=\hat{z})$	M_2	RoM
2	$2O$ (48)	1.000	1.000	0.000	1.000
3	dense	1.500	1.498	0.585	1.732
4	$2T$ (24)	1.000	1.000	0.5585	1.715
5	dense	1.500	1.479	0.585	1.732
6	dense	1.500	1.495	0.584	1.732
7	dense	1.498	1.494	0.585	1.732
8	$2I$ (120)	1.427	1.342	0.5530	1.714
9	dense	1.500	1.497	0.585	1.732
10	dense	1.500	1.491	0.585	1.732

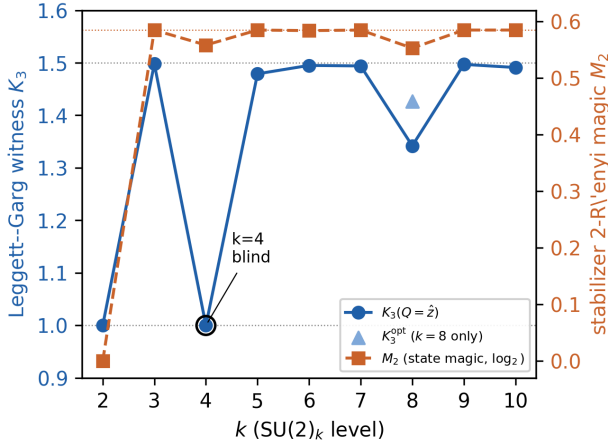


FIG. 1. The $k = 4$ dissociation. Top: Leggett–Garg witness $K_3(Q=\hat{z})$ (blue) against stabilizer 2-Rényi magic M_2 (orange, $\log_2$ basis) across $k = 2, \dots, 10$. At $k = 4$ the witness sits at its macrorealistic ceiling, $K_3 = 1.000$ (open circle, blind), while $M_2 = 0.5585$ is at 95% of the dense-representation ceiling $\log_2(3/2) = 0.585$ (dashed line). At every other finite or dense k the witness exceeds its classical bound and is correspondingly not blind. The $k = 8$ point shows the fixed-axis value $K_3(Q=\hat{z})$; the axis-optimized value $K_3^{\text{opt}} = 1.427$ (Table I) is shown as a lighter marker for comparison.

equal-time-step protocol of Table I ($V_1=V_2$) the $k = 4$ blindness holds for *every* measurement axis, a closed result established for the companion $SU(2)_k$ Leggett–Garg witness [27] (its $M(U)$ eigenvalues exceed 1 only for rotation angles in $(0^\circ, 90^\circ)$, which the tetrahedral group does not contain); under the more general two-propagator pro-

cedure of Sec. IV the blindness is instead bound to alignment, and a misaligned axis reaches $K_3 = 3/2$ — the alignment mechanism itself, not a defect.

Meanwhile, the single-qubit fusion channel of the same braid generator carries near-maximal magic, $M_2 = 0.5585$, i.e. 95% of the finite-dimensional ceiling $\log_2(3/2) = 0.585$ approached by the dense representations, and a robustness of magic RoM = 1.715 close to the dense ceiling $\sqrt{3} = 1.732$. We stress the framing: $K_3 = 1$ at $k = 4$ is a *single-qubit corollary* of a specific braid rotation angle, not a new physical phenomenon, and the finding is that the witness misses a resource that is, by every available magic monotone, almost as large as it can be for this family.

IV. STRUCTURAL THEOREM: ALIGNMENT, NOT FINITENESS OR CLIFFORD

We now explain the $k = 4$ blind spot as a theorem rather than a numerical coincidence. Consider the temporal Leggett–Garg protocol with a fixed dichotomic observable $Q = \hat{n}_0 \cdot \sigma$ (here $\hat{n}_0 = \hat{z}$) and two independent propagators V_1, V_2 drawn from a group $G \subset SU(2)$, giving Heisenberg observables $Q_1 = Q$, $Q_2 = V_1^\dagger Q V_1$, $Q_3 = (V_2 V_1)^\dagger Q (V_2 V_1)$ with Bloch directions $\hat{n}_1 = \hat{n}_0, \hat{n}_2, \hat{n}_3$ in the orbit of $\hat{n}_0$ under the $SO(3)$ image of G . For any single-qubit dichotomic pair, $\{\hat{n} \cdot \sigma, \hat{m} \cdot \sigma\} = 2(\hat{n} \cdot \hat{m})\mathbb{1}$, so the Lüders two-time correlator reduces to a Bloch dot product independent of the state,

$$C_{ij} = \hat{n}_i \cdot \hat{n}_j \quad \text{for every } \rho, \quad (1)$$

and consequently $K_3^{\text{max}} = \max_{(\hat{n}_2, \hat{n}_3) \in \text{orbit}(\hat{n}_1 \cdot \hat{n}_2 + \hat{n}_2 \cdot \hat{n}_3 - \hat{n}_1 \cdot \hat{n}_3)} (\hat{n}_1 \cdot \hat{n}_2 + \hat{n}_2 \cdot \hat{n}_3 - \hat{n}_1 \cdot \hat{n}_3)$ depends only on the Gram matrix of Bloch dot products within the orbit of $\hat{n}_0$ under G — *not* on whether the braid representation is finite, and not on whether G is a Clifford subgroup. The real $k = 4$ braid image is the binary tetrahedral group $2T$ (projective order 12), and σ_1 acts as a projective threefold rotation about $\hat{z}$: the orbit of $\hat{z}$ under the full projective group $2T$ is exactly the four vertices of a tetrahedron, with every off-diagonal Bloch dot product equal to $-1/3$. Exhaustive enumeration over the $4^2 = 16$ orbit pairs (with $\hat{n}_1 = \hat{n}_0$ fixed) gives $K_3^{\text{max}} = 1$ exactly, matching Table I. The same Gram argument applied to the real $k = 8$ braid image (binary icosahedral group $2I$, twelve icosahedral Bloch directions, dots $\{\pm 1, \pm 1/\sqrt{5}\}$) gives $K_3^{\text{max}} = 3/\sqrt{5} = 1.341641$ in the fixed- $\hat{z}$ -axis protocol, and $K_3^{\text{opt}} = 2 \cos 72^\circ - \cos 144^\circ = 1.427$ under the axis-optimized protocol in which the measurement axis is rotated to be perpendicular to a fivefold orbit axis unreachable from the fixed $\hat{z}$ frame — the two $k = 8$ numbers in Table I differ by measurement-protocol scope, not by inconsistency.

The corollary that gives this theorem its bite is a direct counterexample to the plausible-sounding slogan “a finite braid image or a Clifford-generating group is witness-blind”: the chiral octahedral group (24 elements, itself

a genuine single-qubit Clifford subgroup) with the measurement axis *rotated* to $\hat{n}_0 = (1, 1, 0)/\sqrt{2}$ instead of a coordinate axis has orbit dots $\{0, \pm\frac{1}{2}, \pm 1\}$ and reaches $K_3^{\max} = 3/2$ exactly, the full Lüders bound — realized operationally by two genuine $SU(2)$ qubit unitaries reproducing the same value to machine precision. The same finite, Clifford-generating group is witness-blind at one axis and witness-saturating at another; neither finiteness nor the Clifford property is the driver. What is sufficient for blindness is *alignment*: the fixed measurement axis coincides with a sufficiently high-order symmetry axis of the orbit (here, $\hat{z}$ is the threefold axis of the $k = 4$ tetrahedral orbit), forcing the orbit Gram matrix into a form whose maximum is exactly 1. We emphasize, following the explicit abstraction argument, that Gottesman–Knill classical simulability of Clifford circuits is not invoked anywhere in this argument and does not by itself imply Leggett–Garg inertness; the mechanism is the Bloch-sphere Gram geometry of the orbit, evaluated in Eq. (1), not computational-complexity simulability. Correspondingly, the correct statement of the $k = 4$ result is that *this witness measures orbit alignment, not magic or universality*; it says nothing about other witnesses or other measurement axes.

V. THREE CERTIFICATES OF GENUINE NONSTABILIZERNESS

Beyond the single-qubit dissociation of Secs. III–IV, we report three methodologically independent certificates of genuine nonstabilizerlessness, in the same spirit as the exact $SU(2)_k$ mana computation of Fliss [14], the exact-setting anchor we follow most closely among prior anyon-magic constructions. The first two, (1) and (2), certify genuine *long-range* (many-body, topologically encoded) magic not reducible to a local artifact; the third, (3), is the complementary gate-based non-Cliffordness axis of the map (axis (3), Sec. II) — a single-generator quantity included here as a methodologically distinct third line of non-Cliffordness evidence, not a long-range-state certificate in the sense of (1)–(2). Each of the two long-range certificates is a *sufficient* condition for genuine long-range magic (a noninteger witness value, or a gauge-invariant lower bound bounded away from zero); we make no claim that either is *necessary*, and do not rule out other long-range-magic witnesses that these two constructions happen to miss.

(1) *Doubled-Fibonacci mutual-information witness.* Following the long-range-nonstabilizerlessness mutual-information construction of Korbany *et al.* [20], we evaluate the Shannon entropy of the topological-sector weight distribution on the doubled-Fibonacci ground space, $d_{(a,b)} = d_a d_b$ for $(a,b) \in \{1, \tau\} \times \{1, \tau\}$: with quantum dimensions $d = (1, \varphi, \varphi, \varphi^2)$, $\varphi = (1 + \sqrt{5})/2$, the normalized weights are $p = d^2/D^2$ with

$$D^2 = \sum_a d_a^2 = 1 + 2\varphi^2 + \varphi^4 \approx 13.09, \text{ giving}$$

$$H = - \sum_a p_a \log_2 p_a = 1.700979, \quad (2)$$

a noninteger value that certifies genuine long-range magic (Korbany’s Theorem 1: an integer value would indicate a stabilizer ground space and no certificate). We reproduce this value by a second, genuinely independent route: the vacuum column of the doubled-Fibonacci modular S -matrix extracted from a 3×3 lattice (Wilson-loop plus C_3 -rotation channel, with no analytic quantum dimensions or F/R symbols entering the extraction) gives normalized weights $|S_{0a}|^2$ whose Shannon entropy agrees with the direct evaluation to $|\Delta| = 1.8 \times 10^{-11}$, at the level of the lattice-extraction residual. (The closed-form rearrangement $H = \log_2 D^2 - \sum_a p_a \log_2 d_a^2$ is algebraically identical to the direct sum and is retained only as an internal consistency check, not as an independent route.) The extracted vacuum column and its evaluation script are included in the deposit. As a null control on the same witness construction, the $\mathbb{Z}_2$ toric-code ground space (four sectors of equal quantum dimension $d = 1$) gives the integer value $H = 2.000000$ exactly, confirming the witness correctly reports *no* certificate on a stabilizer code; a second null anchor, the binary-entropy toric-code T -state of Korbany’s Eq. (35), $H_2(\cos^2 \frac{\pi}{8}, \sin^2 \frac{\pi}{8}) = 0.600876$, is also reproduced to the digits shown (an earlier candidate value 0.5436 for this same anchor is confirmed incorrect).

(2) *Gauge-invariant minimum ground-space stabilizer Rényi entropy.* A representative-dependent estimate of the stabilizer 2-Rényi entropy on the doubled-Fibonacci ground space gives $M_2 = 6.393$ (Monte Carlo over 1.5×10^5 Pauli samples, standard error 0.13, in the same reduced-Pauli-string-sampling spirit as the many-body SRE estimator of Ding, Wang, and Yan [24]), a robust *lower bound*. Minimizing instead over the full four-dimensional ground space (gauge-invariant construction) gives a range $M_2 \in [6.5, 8.9]$, i.e. $\gg 0$ under every representative, the genuine long-range-magic certificate. As a discriminator, the fraction of Pauli-expectation samples landing exactly in $\{0, \pm 1\}$ (the stabilizer signature) is 100% for the $\mathbb{Z}_2$ toric-code control (where the corresponding $M_2 = 0$ exactly) and only 54% for the doubled-Fibonacci ground space, confirming the qualitative long-range-nonstabilizerlessness diagnostic advocated by Wei and Liu [22] in a related long-range-magic complexity framework, evaluated here on a distinct family (doubled-Fibonacci vs. their surveyed codes and phases); we do not import any specific numeric value from Ref. [22].

(3) *Complementary gate-based non-Cliffordness.* As the gate-based axis of the map (axis (3), Sec. II), distinct from the state magic of (1)–(2), we evaluate the amortized, gate-based stabilizer Rényi entropy of the braid generator itself (rather than of a fixed state), $M^A(\sigma_2)$, across the family: 0, 0.562, 0.765, 0.832, 0.815 for $k = 2, 3, 4, 5, 8$ respectively, with the closed-form single-qubit anchor $M_2^A(T\text{-gate}) = 2 - \log_2 3 = 0.4150$ (agreement

to 3×10^{-15} against three independently constructed anchors: the product state $|0\rangle$, the Bell state, and the tensor product $T|+\rangle \otimes T|+\rangle$, the last matching $2\log_2(4/3)$ exactly). This gate-based measure is nonzero already at $k = 3$ and grows through $k = 5$ before slightly receding at $k = 8$, structurally distinct from the state-magic plateau of Table I and consistent with it throughout (both vanish only at the Clifford point $k = 2$).

VI. HONEST NEGATIVE CONTROL: LEAKAGE-FREE ADDITIVITY

As a negative control accompanying the three certificates above, we verify that a leakage-free construction of the same family remains strictly additive under composition: the construction procedure itself produces no genuine long-range or entangling magic beyond what is already present locally (entanglement contribution $n_{\text{ent}} = 0$; product-state consistency error $\sim 10^{-15}$). On the same construction, the single-qubit Wigner-Mana sanity values across $k = 3, 4, 5, 8$ (base 2, $d = 5$ discrete Wigner representation) are 1.0623, 1.0328, 1.0746, 0.9708, all consistent with the plateau structure of Table I and none suggesting an anomalous long-range contribution beyond the additive baseline; the $k = 2$ construction falls outside the $d = 5$ representation used here and is omitted from this specific check. We report this explicitly as a negative result: it rules out one plausible mechanism for spurious long-range magic (leakage across the composition procedure), it does not itself certify anything positive, and the positive certificates of Sec. V stand independently of it.

VII. HARDWARE-ORIENTED PREDICTION

As a single-qubit anchor for a possible hardware signal-to-noise estimate, the closed-form T -gate value $M_2^A(T) = 2 - \log_2 3 = 0.4150$ (Sec. V, item 3) provides a magic-monotone scale against which a device's discrimination of a magic state from the nearest stabilizer state could be benchmarked. We state this explicitly as a *prediction*, not a hardware result: on real hardware the magic state is not distinguishable from a pure T -gate output by this monotone alone (both saturate the same single-qubit value), and no multi-qubit or topological hardware claim is made here; extending this anchor to a genuine multi-qubit hardware demonstration is a distinct, separately resourced undertaking outside the scope of this paper. Non-Abelian Fibonacci-anyon braiding has already been demonstrated on a superconducting processor [15], but that demonstration addresses universality, not a magic-monotone measurement; closing this gap on real hardware remains open. Universal topological gates have now been demonstrated on qubit hardware for the S_3 quantum double, with universality supplied by fusion as a primitive and a topologically prepared magic state [28] — consistent with the magic-not-braiding resource split

mapped here.

VIII. INTERFACE WITH CONTEXTUALITY AT $d = 3$, AND LIMITATIONS

We map the $k = 4$ dissociation onto an interface with Kochen–Specker–Klyachko (KCBS) contextuality [23, 25] at qutrit dimension $d = 3$, evaluated on the same fusion-channel qutrit used throughout this paper. At $k = 4$ the KCBS witness reaches 2.0595, exceeding the classical bound 2 and certifying contextuality (in contrast to the Leggett–Garg witness of Sec. III, which is blind at the same k); at $k = 3, 5, 8$ the witness reaches 2.22, 2.23, 2.23, approaching the quantum bound $\sqrt{5} = 2.2361$. We make no priority claim on braiding exhibiting KCBS contextuality as such: Chou [23] already reports state-dependent contextuality violations for Fibonacci and $SU(2)_k$ braid representations; the contribution here is the pairing of that witness with the magic monotones of this paper at the same k . Evaluated against the qutrit Wigner-Mana value across the orbit at each of $k = 3, 4, 5, 8$, the Pearson correlation between the KCBS witness and magic is weak and nongeneric ($-0.03, -0.19, -0.07, -0.01$ respectively; consistently negative and, at its largest magnitude ($k = 4$), still only 19% of the way to a perfect correlation), which we read as evidence against a generic one-to-one correspondence between this contextuality witness and magic on this family, evaluated pointwise over a single finite k — not as a claim that no such correspondence could hold more generally, and not in tension with the HWVE theorem discussed in Sec. I for the three reasons given there. The $d \geq 3$ question of whether *any* correlation witness can be proven to always detect magic on this family is left to future work; the present paper contributes only the magic side of that interface.

We close with the limitations of the present scope. First, we do not claim a two-dimensional area-law scaling of long-range magic; established methods do not currently reach system sizes beyond roughly 27 qubits by exact or efficient sampling methods, so no such scaling claim is made here. Second, the hardware prediction of Sec. VII is a single-qubit anchor only; no multi-qubit hardware claim is made. Third, we do not claim a new genuinely temporal magic witness: the naive temporal Leggett–Garg witness studied here is demonstrably blind at $k = 4$ (Secs. III–IV), and constructing a temporal witness that is not subject to the same alignment blind spot is an open problem we do not attempt to solve. Fourth, a doubled-Ising extension of the family studied here (nominal $M_2 \approx 5.3$ under a $\log_3$ convention) is deliberately excluded from the present, deliberately minimal paper; we checked the two most relevant candidate anchors in the literature, the non-Abelian long-range-magic survey of Zhang *et al.* [21] and the long-range-nonstabilizerness complexity framework of Wei and Liu [22], and found neither reports an Ising, $\log_3$, or comparable value, so no prior anchor currently supports including it as a load-

bearing result, and we defer it to future work.

DATA AND CODE AVAILABILITY

The reproduction scripts and raw JSON outputs supporting the values reported in this paper are deposited at [doi:10.5281/zenodo.21134455](https://doi.org/10.5281/zenodo.21134455) and released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license. The deposited outputs include the gauge-invariant ground-space stabilizer-Rényi-entropy range $M_2 \in [6.5, 8.9]$ of Sec. V(2) (`p5a_tsre_robust_m2_range.json`, with its provenance block); the underlying exact-diagonalization construction of the ground space is a separate, heavier computation and is not part of this record.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI,

learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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- [1] V. Veitch, S. A. H. Mousavian, D. Gottesman, and J. Emerson, *The resource theory of stabilizer computation*, New J. Phys. **16**, 013009 (2014); [arXiv:1307.7171](https://arxiv.org/abs/1307.7171).
 - [2] M. Howard and E. Campbell, *Application of a resource theory for magic states to fault-tolerant quantum computing*, Phys. Rev. Lett. **118**, 090501 (2017); [arXiv:1609.07488](https://arxiv.org/abs/1609.07488).
 - [3] R. A. Macedo, P. Andriolo, S. Zamora, D. Poderini, and R. Chaves, *Witnessing magic with Bell inequalities*, [arXiv:2503.18734](https://arxiv.org/abs/2503.18734).
 - [4] S. Cusumano et al., *Non-stabilizerness and violations of CHSH inequalities*, [arXiv:2504.03351](https://arxiv.org/abs/2504.03351).
 - [5] M. Howard and J. Vala, *Nonlocality as a benchmark for universal quantum computation in Ising anyon topological quantum computers*, Phys. Rev. A **85**, 022304 (2012); [arXiv:1112.1516](https://arxiv.org/abs/1112.1516).
 - [6] M. Howard, *Maximum nonlocality and minimum uncertainty using magic states*, Phys. Rev. A **91**, 042103 (2015); [arXiv:1501.05319](https://arxiv.org/abs/1501.05319).
 - [7] M. Howard, J. Wallman, V. Veitch, and J. Emerson, *Contextuality supplies the “magic” for quantum computation*, Nature **510**, 351 (2014); [arXiv:1401.4174](https://arxiv.org/abs/1401.4174).
 - [8] A. J. Leggett and A. Garg, *Quantum mechanics versus macroscopic realism: Is the flux there when nobody looks?*, Phys. Rev. Lett. **54**, 857 (1985).
 - [9] C. Budroni and C. Emary, *Temporal quantum correlations and Leggett-Garg inequalities in multi-level systems*, Phys. Rev. Lett. **113**, 050401 (2014); [arXiv:1309.3678](https://arxiv.org/abs/1309.3678).
 - [10] M. H. Freedman, M. Larsen, and Z. Wang, *A modular functor which is universal for quantum computation*, Commun. Math. Phys. **227**, 605 (2002); [arXiv:quant-ph/0001108](https://arxiv.org/abs/quant-ph/0001108).
 - [11] M. Freedman, M. Larsen, and Z. Wang, *The two-eigenvalue problem and density of Jones representation of braid groups*, Commun. Math. Phys. **228**, 177 (2002); [arXiv:math/0103200](https://arxiv.org/abs/math/0103200).
 - [12] G. Kuperberg, *Denseness and Zariski denseness of Jones braid representations*, Geom. Topol. **15**, 11 (2011); [arXiv:0909.1881](https://arxiv.org/abs/0909.1881).
 - [13] L. Byles, E. Forbes, and J. K. Pachos, *Demonstration of magic state power of $D(S_3)$ anyons with two qudits*, [arXiv:2408.03377](https://arxiv.org/abs/2408.03377).
 - [14] J. R. Fliss, *Knots, links, and long-range magic*, J. High Energy Phys. **04**, 090 (2021); [arXiv:2011.01962](https://arxiv.org/abs/2011.01962).

- [15] S. Xu *et al.*, *Non-Abelian braiding of Fibonacci anyons with a superconducting processor*, [arXiv:2404.00091](#).
- [16] C. E. P. Robin and M. J. Savage, *Anti-flatness and non-local magic in two-particle scattering processes*, [arXiv:2510.23426](#).
- [17] L. Leone, S. F. E. Oliviero, and A. Hamma, *Stabilizer Rényi entropy*, *Phys. Rev. Lett.* **128**, 050402 (2022); [arXiv:2106.12587](#).
- [18] L. Leone and L. Bittel, *Stabilizer entropies are monotones for magic-state resource theory*, *Phys. Rev. A* **110**, L040403 (2024); [arXiv:2404.11652](#).
- [19] T. Haug and M. S. Kim, *Scalable measures of magic resource for quantum computers*, *PRX Quantum* **4**, 010301 (2023); [arXiv:2204.10061](#).
- [20] D. A. Korbany, T. D. Ellison, D. T. Stephen, and L. Piroli, *Long-range nonstabilizerness of topologically encoded states from mutual information*, [arXiv:2605.22424](#).
- [21] Y. Zhang, I. H. Kim, Y. Bao, and S. Vijay, *Extensive long-range magic in non-Abelian topological orders*, [arXiv:2605.15150](#).
- [22] F. Wei and Z.-W. Liu, *Long-range nonstabilizerness and quantum codes, phases, and complexity*, [arXiv:2503.04566](#).
- [23] T.-M. Chou, *Topological contextuality and quantum representations*, [arXiv:2506.14537](#).
- [24] Y.-M. Ding, Z. Wang, and Z. Yan, *Evaluating many-body stabilizer Rényi entropy by sampling reduced Pauli strings*, [arXiv:2501.12146](#).
- [25] A. A. Klyachko, M. A. Can, S. Binicioğlu, and A. S. Shumovsky, *Simple test for hidden variables in spin-1 systems*, *Phys. Rev. Lett.* **101**, 020403 (2008).
- [26] B. Y. Sayim, *Bell Algebra without Bell Violation: CHSH Operators for Ising Anyons in the Kitaev Honeycomb Model*, Zenodo preprint (2026); [doi:10.5281/zenodo.21134323](#).
- [27] B. Y. Sayim, *A dichotomous Leggett–Garg witness for braid-representation density in $SU(2)_k$: exact at $d = 2$, dimension-limited at $d \geq 3$* , Zenodo preprint (2026); [doi:10.5281/zenodo.20531124](#) [Concept-DOI, always points to latest version].
- [28] C. F. B. Lo, A. Lyons, D. Gresh, M. Mills, P. E. Siegfried, M. D. Urmey, N. Tantivasadakarn, H. Dreyer, A. Vishwanath, R. Verresen, and M. Iqbal, *Universal gates from braiding and fusing anyons on quantum hardware*, *Nature* **655**, 591 (2026); [arXiv:2601.20956](#).